

Make a PDF file with your name <FirstName_LastName>. The file should have the following information, and must be updated after every presentation.

Presentation 5

- Links to presentation and code on Github
 - Code
 - https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/Outcome_Tagging_Dec8_Derwent.ipynb
 - Presentation
 - https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/CallOutcomesFinal_9Dec_EDA2.pdf
 - Presentation video
 - https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/VideoLink_Nov5_EDASimile2.txt
- What did you do?
 - For the final presentation, I continued working on the outcome classification and made sure that all the datapoints were being captured and tagged correctly.
 - I then made stacked bar charts that broke down the proportions of negative, neutral, and positive outcomes for the caller types (determined by Krish). I also made a stacked bar chart for the different subtypes (reasons a call ended) for the caller types.
- How does it help the project?
 - Knowing the distribution of call outcomes and the reason for the call outcomes can give great insight on what issues are being served and what issues need more resource allocation. For example, for non-senior legal callers, there is a very high rate of closed queue or abandonment calls that greatly outweigh the number of successful calls. I would recommend that LegalAid focuses on this menu to see if they can allocate more people to answer these calls so that more people can be served.
 - The visualizations are very helpful in getting a general idea of which callers are being serviced and which ones need more attention. Then, the tables give you a detailed and precise look at the results.
- Issues Faced
 - I attempted to tag all of the calls, but afterwards there were ~66,000 calls that did not have any outcome tagged. I could not figure out why these calls were not getting tagged.
- Attempts to Resolve Issues

- I checked with Vivienne to see if our outcome classification was different, and I made sure to have the same criteria as her when classifying. I realized that I was missing some special conditions that required more than just the last activity name, so I tried to implement her code. I was unable to synthesize her code and mine.
- Next Steps
 - Since this is the last presentation, my recommendation to Cynthia would be to look at which menus are largely being abandoned (ex: FarmersMenu) and add more people that can serve the call there
 - Additionally, I would offer the option to schedule a courtesy callback even if the queue is closed so that a client doesn't have to call multiple times, reach a closed queue menu, and have to schedule a callback on their perhaps third or fourth call.
- References
 - Krish's code for caller type tag from December 1
 - I built off this code as Krish had categorized the calls into specific caller types. I tagged all of the calls and visualized the results according to these categories.
 - Vivienne and I also worked together to tag the calls and check each other's work. As mentioned above, my code is missing ~66,000 calls and I was unable to find where these calls were being lost, so for our presentation we used the numbers that Vivienne calculated. Her numbers are more accurate. Since we wanted to get the most accurate numbers, I shared my code that I used to make the graphs with her so that we could visualize all of the call outcomes.

Presentation 4

- Links to presentation and code on Github
 - Code
 - https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/Consistency_Check_CallOutcome_Nov17_Derwent.ipynb
 - Call Outcomes Excel Sheet
 - https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Clarifications/CAR_data_Activity_name_tagging_10Nov.xlsx
 - Presentation
 - https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/OutcomeTagging_Nov18_EDASimple2.pdf
 - Presentation Video

- https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/VideoLink_Nov18_EDASample2.txt

- What did you do?
 - I worked on classifying and verifying the call outcomes on the excel spreadsheet. I mainly looked at the flowchart and the verbiage sheet to classify a call's last activity. For most activities, the judgement was based on common sense. For example, if a call's last activity was in a menu page and we expect it to reach a queue/agent then I would class that call as negative and abandoned.
 - To verify, I made sure that I was getting the same number of calls per last activity as Krish, then I looked at each activity name and took a look at a few calls to get a general sense that the classification was accurate. For some activities, I was more unsure of what outcome to assign, so I looked at more calls within the category.
 - There were some outcomes that I updated (ClinicVoicemailTransfer and DisconnectContact1) to capture more scenarios.
- How does it help the project?
 - The classification of calls as positive, negative, or neutral helps LegalAid see where they are succeeding in helping their clientele and where they could devote more resources to. This can also help see if certain menu prompts are confusing or frustrating, as many people are abandoning their call and not getting served. Having a baseline categorization helps all teams determine if clients are being served or not.
- Issues Faced
 - There were just a few activity names that we were confused on its meaning:
 - GetLogged<>Agents – this appears before an agent logs on to the call, so we were wondering if this was a system error or if the client abandoned the call before they could speak with the agent
 - TransfertoSafeHaven – we are unsure which step in the family menu the client must push to access the safe haven transfer, its node is not noted in the flow chart. Looking at the calls with this last activity I saw that all of them are transferred to the CBT Agent, so we assumed that this would be a positive outcome
 - LegalServerScreenPop & ScreenPopProcessComplete – this typically appears before the customer talks to an agent, but we were unsure of how to classify when this is the last step. We assumed that this was a system error where the customer was supposed to be connected with an agent, but the call fell through.
 - No Activity – unable to find this in data
- Attempts to Resolve Issues

- For this presentation we made assumptions that are detailed above in our outcome classification. I looked at calls with these activities names to try to discern an outcome, so what is in the excel sheet is my best guess of these outcomes.
- Next Steps
 - I want to make sure to get some clarification on the activity names that were ambiguous
 - Otherwise we will take the call outcome analysis and dive deeper into how to turn negative calls into positive calls
- References
 - N/A

Presentation 3

- Links to presentation and code on Github
 - Code
 - https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/Outcome_Classification_22Oct_Derwent.ipynb
 - Presentation
 - https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/CallMenuOutcomes_4Nov_EDATeam2.pdf
 - Presentation Video
 - https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/VideoLink_Nov5_EDASample2.txt
- What did you do?
 - For this presentation, I built off our objective in the last presentation to tag the calls in the CAR dataset as positive, neutral, or negative. As discussed with Krish, we decided to split up the calls by menu and tackle the 5 most common menus that clients navigate towards. I worked on the housing menu. In this menu, there are several queues that a customer can be placed in if the agent is busy (Consumer, Housing, or Clinic Voicemail Transfer). I looked at calls that reached these queues, as well as all calls that navigated to the housing menu.
 - There were 289 calls that navigated through the housing menu. I created a function that assigned tags based on whether there was a certain termination reason (these categories were determined in presentation 2). I then applied it to all the calls in the housing menu and counted the number of outcomes. It was ~80% negative and ~20% positive, with no calls being recorded as neutral.

- One thing to note in this analysis is that I automatically labeled “Customer Left” as a negative outcome unless in the call there was an agent name assigned with the call. I know that there are cases that could be considered positive or neutral, but since the Simple EDA Team 1 is focusing on abandoned calls, I decided to incorporate their analysis for the next presentation. Therefore, these numbers may be slightly skewed towards the negative.
- Then, I specifically looked at calls that navigated to the housing menu and reached either the Consumer Queue, Housing Queue, or the Client Voicemail Transfer. For these calls, I considered it a successful outcome if the customer was able to speak to an agent and any other outcomes as negative. A key takeaway from this is that once a customer reaches a queue and requests a callback, the customer does get help in most cases within the housing queues.
- How does it help the project?
 - This analysis helps the project because it can see if LegalAid is able to serve most of the calls that they receive and in which menu areas/queues they need more resources in to be able to serve more calls. In the housing menu, it seems that there are not as many callers, and most callers in a queue are served so the resource allocation for this menu is successful.
- Issues Faced
 - This was an uncertainty as I know that Simple EDA Team 1 has been working a lot with classifying abandoned calls, so I did not do as much of an indepth analysis on the ‘customer left’ termination reasons.
 - Additionally, in the Clinic Voicemail Transfer menu, all of the calls are classified as positive because there is ‘CBT Agent’ assigned to each call. This is the chatbot agent, but I am still uncertain if I can confidently say that every customer that reaches the chatbot is able to be served. This is something that I will ask Cynthia and the Simple EDA Team 1.
- Attempts to resolve issues
 - I made an assumption that if a person was being served by the CBT Agent (the chatbot) that this would be a positive outcome. However, this is something that I will ask Cynthia as I do not know if this is always the case.
- Next Steps
 - Since EDA team 1 is also currently working on abandoned calls, I will implement how they classified their calls into my analysis on the housing menu once they are finished.
- References
 - I did not use her code, but to identify positive call outcomes for calls that went to a queue I used Sophia Jukovich’s idea that a positive outcome is if a courtesy callback is attached to an agent name in the activity.

Presentation 2

- Links to presentation(s) and code(s) on GitHub
 - Code:
 - https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/Outcome_Classification_22Oct_Derwent.ipynb
 - Presentation
 - https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/CallCategorization_Oct23_Group2.pdf
- What did you do?
 - I, along with Vivienne, helped make definitions for when call outcomes are positive, neutral, or negative. I looked at CAR dataset to see what journeys clients were taking and how their calls were terminating. After creating some definitions based on logic and some case studies, I started sorting the termination reasons into positive, neutral, and negative. There are some termination reasons (like 'Customer Left') that could be classified in multiple ways. For my analysis, I focused more closely on 'Queue Timeout' as I initially thought it was a negative outcome, but I then looked more closely and noticed that many of calls ending with this queue timeout also receive a courtesy callback, so there are a few ways that a queue timeout can be classified.
- How does it help the project?
 - This analysis helps the project, specifically with the courtesy callbacks, as I am able to see how many clients are being served with the callbacks, as well as how many times an agent has to call the client back before the client responds. Trying to identify good times to give a call back and allocating resources for agents performing the callbacks could be beneficial so that the agent does not have to call the client multiple times to try to reach them, as all of these calls do not serve any purpose.
- Issues faced (if any)
 - I found a call that was a callback and the second to last step in the journey was "agent left", so I classified this call as positive. However, the last row in the Contact Session ID showed the caller being in the queue "Family" for 2 minutes after the agent left. I was unsure of what this meant and will need to ask Cynthia/Krish for more clarification.
- Attempts to resolve issues (if any)
 - Will ask Krish/Cynthia in mentor meetings
- Issues resolved (if any)
- Next steps

- My next steps are to start systematically tagging calls with positive, neutral, or negative outcomes. For this presentation, I did not code a lot and did a more exploratory approach. For the next presentation, I will encode the findings I have so that all the calls in the dataset can be tagged, not just the ones I used for my case studies.
- References (Mention if you built up on someone else's work)
 - N/A

Presentation 1

- Links to presentation(s) and code(s) on GitHub
 - Presentation
 - https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Data%20integrity%20checks/CARAllCallsIntegrity_7Oct_EDATeam2.pdf
 - Code
 - https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Internal%20work%20of%20Teams/EDA_simple_2/CAR_task2_pres1_Derwent%20copy.ipynb
- What did you do?
 - For this presentation, I tried to get myself familiar with the CAR dataset by manually exploring the different journeys a call can take. My goal is to be able to systematically track calls and find which menus callers are lingering on/ultimately trying to reach. I mainly looked for when a customer is reaching a queue (either the Queue Times out or they reach a Closed Queue) to see which queues are being accessed the most.
- How does it help the project?
 - I believe that tracking which queues are trying to be reached can be helpful with support distribution. Due to the volume of calls, it is inevitable that many callers will reach a queue, so I wanted to see which queues should be prioritized with more staff.
- Issues faced (if any)
 - I noticed some inconsistencies within the data - mainly that there are several rows within one contact session that list a termination reason. Initially, I assumed that the termination reason would be the last row of a call as there is no other actions that can be taken after this, but I found that there can be multiple rows for a call that have a Termination Reason. My next goal is to figure out what these mean and whether the multiple rows are simply giving duplicate information, or if I need to make my search for queue names more complex.

- Attempts to resolve issues (if any)

- This is something that I will ask Krish during our mentor meetings.

- Issues resolved (if any)

- Next steps

- My next steps is to code a more systematic way of tracking queue placements and menu selections after clarifications from Krish.

- References (Mention if you built up on someone else's work)

- N/A